



EUNOIA JUNIOR COLLEGE

JC1 Promotional Examination 2018

General Certificate of Education Advanced Level

Higher 1

TYPESETTING TIPS UNVEILED

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Paper 1 [100 marks]

12 September 2018

2 hours 30 minutes

Additional Materials: Answer Paper

List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and question number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Q1

Coefficient of x^2 is negative, for $(k-8)x+2k-2x^2$ is always negative, discriminant $b^2-4ac < 0$

$$\begin{aligned}b^2-4ac &= (k-8)^2 - 4(-2)(2k) \\&= k^2 - 16k + 64 + 16k \\&= k^2 + 64 > 0 \text{ for all real values of } k\end{aligned}$$

Therefore discriminant will never be less than 0 and there are not real values of k for $(k-8)x+2k-2x^2$ is always negative

Q2

Let r , b and y be the number of red, blue and yellow matchsticks respectively.

$$r + b + y = 100 \text{ --- (1)}$$

$$\frac{y}{5} + r = b \Rightarrow 5r - 5b + y = 0 \text{ --- (2)}$$

$$\frac{b}{3} + \frac{y}{5} = 4\left(\frac{r}{4}\right) \Rightarrow 15r - 5b - 3y = 0 \text{ --- (3)}$$

Using GC, $r = 20$, $b = 30$ and $y = 50$

5 squares, 10 triangles and 10 pentagons formed.

Q3

(i) Let the length of the rectangle be y m.

$$\text{Perimeter} = 40\pi$$

$$40\pi = 2\pi(2x) + 4x + 2y$$

$$y = 20\pi - 2\pi x - 2x$$

Let the area of the flower-bed be A .

$$A = \pi(2x)^2 + 2xy$$

$$A = \pi(2x)^2 + 2x(20\pi - 2\pi x - 2x)$$

$$A = 40\pi x - 4x^2$$

$$A = 4x(10\pi - x) \quad (\text{shown})$$

(ii)
$$A = 40\pi x - 4x^2$$

$$\frac{dA}{dx} = 40\pi - 8x$$

$$\frac{dA}{dx} = 0$$

$$40\pi - 8x = 0$$

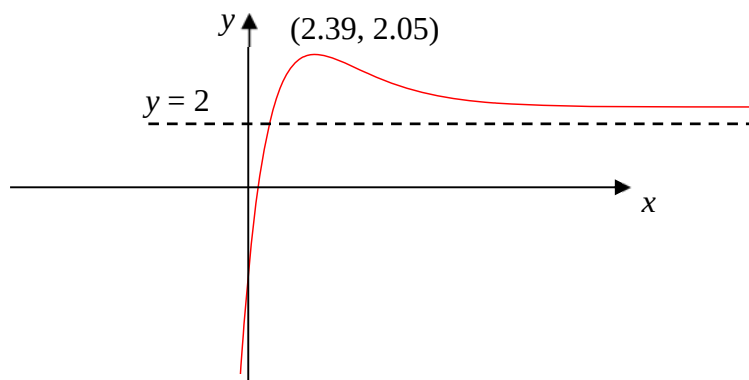
$$\Rightarrow x = 5\pi$$

$$\frac{d^2A}{dx^2} = -8 < 0$$

Hence $A = 40\pi(5\pi) - 4(5\pi)^2 = 100\pi^2$ is a maximum value when $x = 5\pi$.

Q4

(i)



$$y = 2 - 2e^{-2x+1} + e^{-x}.$$

$$y = 2 - 2e^{-2x+1} + e^{-x}$$

$$(ii) \frac{dy}{dx} = 4e^{-2x+1} - e^{-x}$$

$$\text{At } x=1, \quad \frac{dy}{dx} = 4e^{-1} - e^{-1} = \frac{3}{e}$$

equation of the tangent to the curve at the point where $x=1$,

$$y - y_1 = \frac{3}{e}(x - x_1)$$

$$y - \left(2 - \frac{1}{e}\right) = \frac{3}{e}(x - 1)$$

$$ey - (2e - 1) = 3(x - 1)$$

$$-3x + ey = 2e - 4$$

$$a = -3, \quad b = -4$$

$$(i) \quad \text{Area bounded} = \int_1^3 2 - 2e^{-2x+1} + e^{-x} \, dx$$

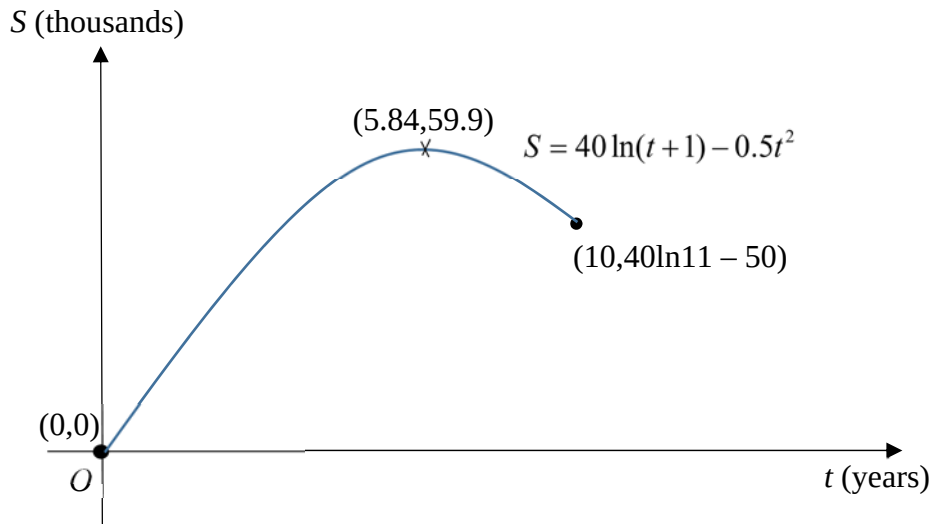
$$\left[2x - \frac{2e^{-2x+1}}{-2} + \frac{e^{-x}}{-1} \right]_1^3$$

$$= \left[2x + e^{-2x+1} - e^{-x} \right]_1^3$$

$$= \left[2(3) + e^{-2(3)+1} - e^{-(3)} - \left(2(1) + e^{-2(1)+1} - e^{-(1)} \right) \right]$$

$$= 4 + \frac{1}{e^5} - \frac{1}{e^3}$$

Q5 (i)



(ii)

$$\frac{dS}{dt} = \frac{40}{t+1} - t$$

$$\frac{dS}{dt} = 0$$

$$\frac{40}{t+1} - t = 0$$

$$40 - t(t+1) = 0$$

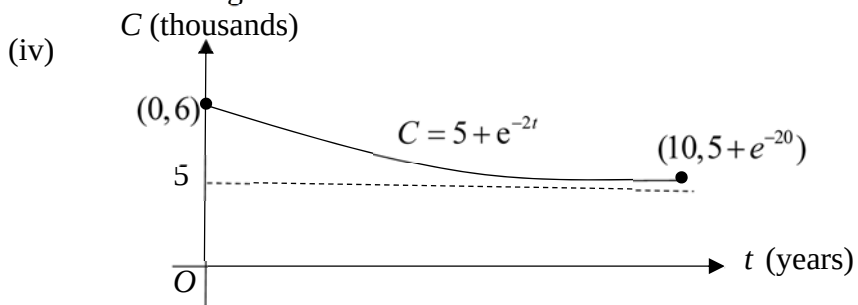
$$t^2 + t - 40 = 0$$

$$t = 5.84 \quad \text{or} \quad t = -6.84 \text{ (reject)}$$

$$\frac{d^2S}{dt^2} = -\frac{40}{(t+1)^2} - 1 < 0$$

Hence at $t = 5.84$, we will have max revenue, the year is 2015.

(iii) The annual revenue would start to decrease after a period of time because with discoveries of new technology, the current LED produced may be outdated and hence causing the decline in sales and thus revenue.



(v) Required area

$$\begin{aligned} &= \int_0^4 5 + e^{-2t} dt \\ &= \left[5t - \frac{1}{2} e^{-2t} \right]_0^4 \\ &= 20 - \frac{1}{2} e^{-8} + \frac{1}{2} e^0 = \frac{41}{2} - \frac{1}{2} e^{-8} \end{aligned}$$

The area found is the total cost incurred over the 4 years, which is \$20500.

Q6 Let X be the random variable denoting the Body Mass Index of a randomly selected adult male i.e. $X \sim N(\mu, \sigma^2)$.

Given: $P(X \geq 25) = 0.362$

$$P\left(Z \geq \frac{25 - \mu}{\sigma}\right) = 0.362$$

$$\therefore \frac{25 - \mu}{\sigma} \approx 0.353118$$

$$25 \approx \mu + 0.353118\sigma \quad \text{-----(1)}$$

Given: $P(X > 30) = 0.055$

$$P\left(Z > \frac{30 - \mu}{\sigma}\right) = 0.055$$

$$\therefore \frac{30 - \mu}{\sigma} \approx 1.59819$$

$$30 \approx \mu + 1.59819\sigma \quad \text{---(2)}$$

Solving (1) and (2):

$$\mu = 23.582 \approx 23.6$$

$$\sigma = 4.0158 \approx 4.02$$

$$\sigma^2 = 16.127 \approx 16.1$$

Q7

(i)

Number of 6-letter code that can be formed using the word PASSES if the letters can be repeated
 $= 4^6 = 4096$ ways

The letters that can be used are P, A, S, E.

P, A, S, E	P, A, S, E	P, A, S, E	P, A, S, E	P, A, S, E	P, A, S, E
4 ways	4 ways	4 ways	4 ways	4 ways	4 ways

(ii) Applying **Grouping Technique** and count SSS as 1 unit. Then, we have 4 units to arrange.

(i) No of 6-letter code that can be formed using the word PASSES if the letters are not repeated and all S's are consecutive $= 4 \times 1 = 24$ ways

(iii)

(ii) No of 6-letter code that can be formed using the word PASSES if the letters are not repeated and first is S and last is E

We are left with P, A, S, S to arrange only since the first position is filled by S and last is E.

Method 1 (Listing)

The 2 S's are side by side:

PASS, PSSA, SSPA, APSS, ASSP, SSAP (6 ways)

The 2 S's are not side by side

PSAS, ASPS, SPSA, SASP, SPAS, SAPS (6 ways)

Total no. of ways = 12

Method 2 (chance to intro them permutation involving identical objects)

$$\text{No. of ways} = \frac{4!}{2!} = 12$$

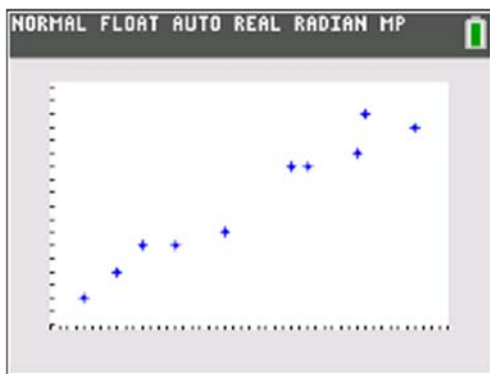
(iv) S S S
 ↑ ↑ ↑ ↑

Insertion method would be useful in separating. We arrange the S, S, S (1 Way) **AND** there are 4 slots to insert P, A, E **AND** following that we arrange the letters P, A, E..

$$\text{No. of ways} = {}^4C_3 \times 3! = 24$$

Q8

(i)

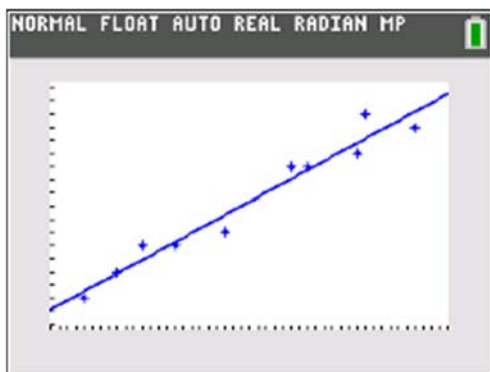


$$r = 0.97693 = 0.977 \text{ (3sfs)}$$

(ii)



$$y = 45.132 + 0.34135x = 45.1 + 0.341x$$



(iii) Since x is the independent variable, y on x should be used for the estimation.

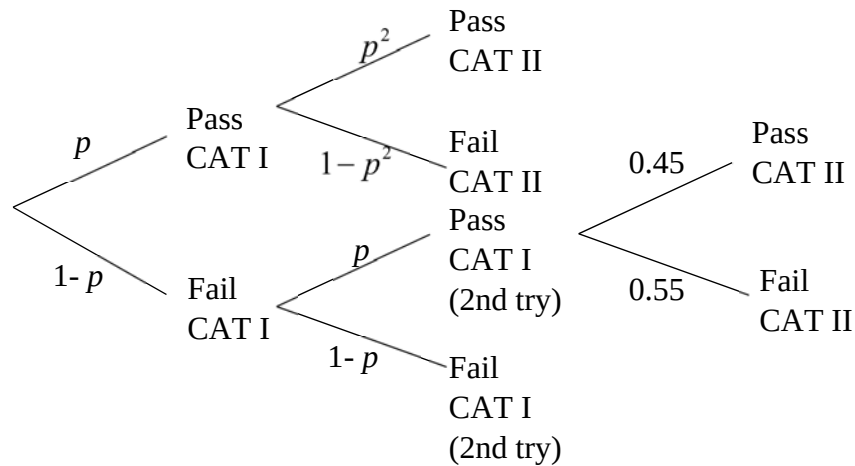
For $x = 70$,

$$y = 45.132 + 0.34135(70) = 69.0265 = 69$$

This estimate is reliable as $x = 70$ is within the given data range and r is close to 1 which indicates a strong positive linear correlation between x and y .

(v) a represent the marks obtained when a student did not do any optional homework problem.

Q9 (i)



(ii) (ii) Required probability $= p(p^2) + (1-p)(p)(0.45) = 0.55$

$$p^3 - 0.45p^2 + 0.45p = 0.55$$

$$p^3 - 0.45p^2 + 0.45p - 0.55 = 0$$

$$p = 0.77887 = 0.779$$

(iii) Required probability

$= P(\text{is successful in the university qualification} | \text{failed CAT I in the first attempt})$

$= \frac{P(\text{is successful in the university qualification and failed CAT I in the first attempt})}{P(\text{failed CAT I in the first attempt})}$

$$= \frac{(1-p)(p)(0.45)}{1-p}$$

$$= 0.45p = 0.35049 = 0.350 \quad (\text{to 3 s.f.})$$

Q10

(i) Assumptions:

1) The event that a balloon is faulty is independent of any other balloon.

2) The probability that a balloon is faulty remains constant throughout the sample.

(ii) Let X be the no. of faulty balloons out of 50.

$$X : B(50, 0.04)$$

$$E(X) = 50(0.04) = 2$$

(iii) $P(X > 2) = 1 - P(X \leq 2) = 0.32329 = 0.323 \quad (\text{to 3 s.f.})$

(iv) Let Y be the no. of packs with more than 2 faulty balloons (out of 20).

$$Y : B(20, 0.32329)$$

$$P(Y < 0.25(20)) = P(Y < 5) = P(Y \leq 4) = 0.17464 = 0.175$$

(v)

$$P(3 \leq Y < 10) = P(Y \leq 9) - P(Y \leq 2) = 0.90161 = 0.902$$

Q11

Let X and Y denote the duration of a randomly chosen call to City J and City C respectively.

$$X \sim N(8, 1.5^2) \quad Y \sim N(10, 1.8^2)$$

(i) $P(X > 11) = 0.02275 = 0.0228$ (3sf)

(ii) Required Probability = $\left[P(X < 11) \right]^3 = (1 - 0.02275)^3 = 0.933$ (3sf)

(iii) The cases in part (ii) is a subset of the cases in part (iii). Therefore, p is greater.

(iv) $X_1 + X_2 + X_3 + Y_1 + Y_2 \sim N(44, 13.23)$

$$P(X_1 + X_2 + X_3 + Y_1 + Y_2 > 48) = 0.135728 = 0.136$$
 (3s.f)

(v) $J \sim N(0.20 \times 8, 0.20^2 \times 1.5^2)$, $C \sim N(0.30 \times 10, 0.30^2 \times 1.8^2)$
 $J \sim N(1.6, 0.09)$, $C \sim N(3, 0.2916)$

$$J_1 + J_2 + J_3 - 2C \sim N(1.6 \times 3 - 2 \times 3, 3 \times 0.09 + 2^2 \times 0.2916)$$

$$J_1 + J_2 + J_3 - 2C \sim N(-1.2, 1.4364)$$

$$P(J_1 + J_2 + J_3 - 2C > 1.50) = 0.0121$$
 (3 s.f)

This is the probability that Ashton will pay more than \$1.50 for the total costs of 3 calls to City J to exceed twice that the cost of one call to City C .

Q12

(i) Let X be the lifetime of a “X-phone” battery and μ denotes the population mean lifetime of a battery in hour.

$$H_0: \mu = 150$$

$$H_1: \mu < 150$$

(vi) $\bar{x} = \frac{\sum x}{n} = \frac{10\,317}{70} = 147.38571$

$$s^2 = \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right] = \frac{1}{69} \left[1\,540\,231 - \frac{(10\,317)^2}{70} \right] = 284.82008$$

To test

$$H_0: \mu = 150$$

$$H_1: \mu < 150$$

Perform one-tailed test at 10% level of significance,

Under H_0 , since $n = 70$ is large, by Central limit theorem,

$$\bar{X} : N\left(150, \frac{284.82}{70}\right) \text{ approximately}$$

$$\text{Test statistic } Z = \frac{147.3857 - 150}{\sqrt{\frac{284.82}{70}}}$$

$p = 0.09748 < 0.10$, we reject H_0 and conclude that at the 10% level of significance, there is sufficient evidence to say that the population mean lifetime of the batteries is less than 150 hours.

(vii) **There is a probability of 0.09748 of obtaining a sample mean lifetime of less than 147.3857 hours from a sample of 70 batteries, assuming that the population mean lifetime of the batteries is 150 hours.**

Or The p-value is the lowest significance level at which the sample mean lifetime of the batteries is less than 150 hours.

(viii) It is NOT necessary, since the sample size $n = 70$ is sufficiently large, the **sample mean** lifetime of the batteries will be normally distributed by the Central Limit Theorem.

$$H_0: \mu = 150$$

$$H_1: \mu \neq 150$$

Level of significance: 10%

Under H_0 , since $n = 70$ is large, by Central limit theorem, $\bar{X} : N\left(150, \frac{15^2}{70}\right)$ approximately.

$$\text{Test statistic } Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{t - 150}{\frac{15}{\sqrt{70}}}$$

Rejection region: Reject H_0 if $z \leq -1.6449$ or $z \geq 1.6449$

Since there is sufficient evidence, at 10% level significance to conclude that the mean lifetime of the batteries has changed,

$$\frac{t - 150}{\frac{15}{\sqrt{70}}} \leq -1.6449 \quad \text{or} \quad \frac{t - 150}{\frac{15}{\sqrt{70}}} \geq 1.6449$$

$$\Rightarrow t - 150 \leq -2.94905 \quad \text{or} \quad t - 150 \geq 2.94905$$

$$t \leq 147.1 \quad \text{or} \quad t \geq 152.9$$

Set of values of t is $\{t \in \mathbb{R} : t \leq 141.7 \text{ or } t \geq 152.9\}$